

Preserving Molecular Information During Pathology Report De-identification: A Synthetic Benchmark

Background: De-identified pathology reports could support model-assisted standardization, summarization, hypothesis generation and testing, and research abstraction. Preparing reports for these uses requires removing identifiers while preserving molecular findings.

Design: We compared an institution-developed pipeline, context-aware rules, and refined context-aware rules with five frontier LLM configurations (GPT-5.6 Sol through Enterprise and a shared model-access interface; Claude Sonnet 5, Gemini 3.1 Pro, and Grok 4.6 through that same interface). Synthetic report pairs tested molecular-expression preservation and removal of similarly formatted identifying digits (200 pairs). A separate experiment used investigator-designed report layouts targeting known weaknesses and contextual challenges (eight patterns; 25 pairs each). Exploratory analyses compared paired outcomes and grouped reports sharing patterns.

Results: LLM configurations achieved both objectives more consistently than the institution-developed pipeline or either rule method (197-200/200 versus 0/200, 133/200, and 156/200, respectively). Molecular preservation was complete with LLMs, incomplete with both rule methods, and absent with the pipeline (200/200, 159/200, and 0/200). Residual identifier exposure was uncommon with refined rules and LLMs (3/200 and 0-3/200). Pair-level comparisons favored LLMs, but pattern-grouped analyses were inconclusive (adjusted $p < .001$; $p = .156$ versus earlier rules and $.375$ versus refined rules). Local failures included molecular over-redaction and identifiers missed across line breaks, separated footers, parentheses, and modified labels. In the separate experiment, refinement halved identifier exposures, with parenthetical and modified-label failures persisting (100/200 to 50/200).

Conclusions: Local refinement improved identifier removal without resolving molecular information loss. LLM configurations performed better descriptively; safety and performance on independent clinical reports remain unestablished.

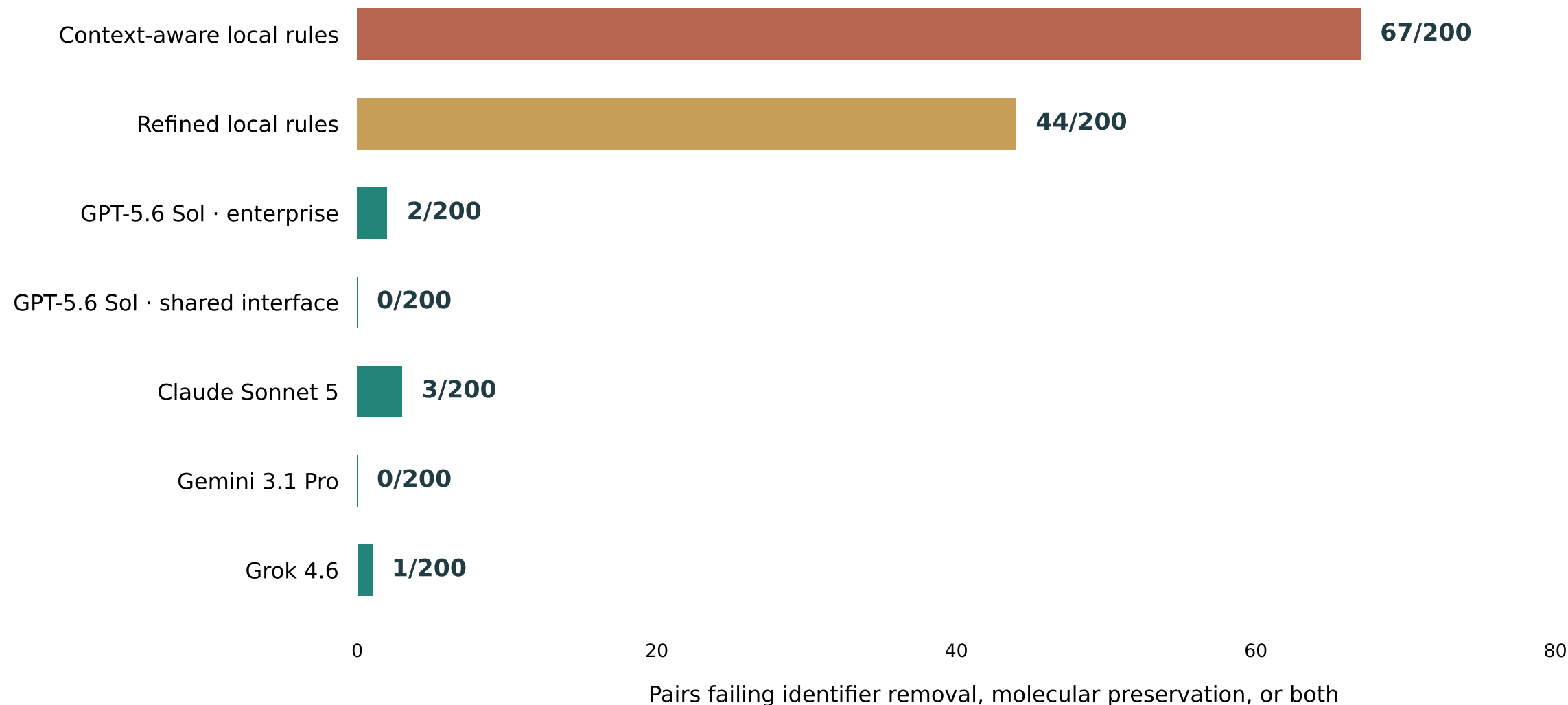


Figure 1. Overall failure in the matched comparison. Each approach processed the same 200 unique synthetic pairs (400 reports). Failure means loss of the tested molecular expression, exposure of the designated identifying digits, or both. Earlier context-aware rules failed 67 pairs, refined rules 44, and LLM configurations 0-3. The institution-developed pipeline failed all 200 pairs (Figure S1). Exploratory pair-level comparisons favored LLMs (adjusted $p < .001$); grouping reports sharing a pattern yielded inconclusive results (adjusted $p = .156$ versus earlier rules; $.375$ versus refined rules). The refined comparison was added subsequently. GPT Enterprise used the chat interface; non-Enterprise GPT, Sonnet, Gemini, and Grok used the same shared model-access interface.

Original synthetic report

Patient name: Lena Varnell
Medical record number: 48271639
Date of birth: 06/18/1965
Ordering provider: Mira Norvale, MD
SURGICAL PATHOLOGY REPORT
Accession: SP32-041728
Collected: 09/10/2032 09:25
Received: 09/10/2032 10:12
Reported: 09/12/2032 15:40

SPECIMEN
A. Right upper lobe lung, image-guided core biopsy

FINAL DIAGNOSIS
Lung, right upper lobe, core biopsy:
Pulmonary adenocarcinoma.

MICROSCOPIC DESCRIPTION
Core fragments contain infiltrating malignant glands with focal papillary architecture, nuclear enlargement, and conspicuous nucleoli. Focal necrosis is present. No definitive squamous differentiation is identified in the sampled tissue.

IMMUNOHISTOCHEMISTRY
TTF-1: Positive in tumor nuclei.
Napsin A: Positive in tumor cytoplasm.
p40: Negative in tumor cells.
The findings support pulmonary adenocarcinoma.

CLINICAL HISTORY
Known EGFR-mutated lung adenocarcinoma. Repeat biopsy for molecular reassessment; correlate with prior therapy and testing.

GROSS DESCRIPTION
Four tan-white cores, 0.3-0.9 cm in length and up to 0.1 cm in diameter; submitted entirely in cassette A1.
Fixative: 10% neutral buffered formalin.

MOLECULAR TESTING
Block A1 selected; estimated viable tumor cellularity 40%.
Targeted DNA sequencing reported separately as MP32-006381.
The diagnosis is based on this biopsy; pathologic stage is not assigned.

Electronically signed: Arlen Voss, MD
Lena Varnell | 48271639 | SP32-041728 | Page 1 of 1

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MOLECULAR PATHOLOGY REPORT
Accession: MP32-006381
Parent accession: SP32-041728
Collected: 09/10/2032 09:25
Received: 09/13/2032 11:05
Reported: 09/18/2032 14:20

SPECIMEN INFORMATION
Site: Right upper lobe lung
Diagnosis: Pulmonary adenocarcinoma
Material: Formalin-fixed paraffin-embedded core biopsy
Specimen part: A | Block: A1
Tumor cellularity: 40% | Macrodissection: Performed

MOLECULAR RESULTS
EGFR p.L858R detected; variant allele fraction 28%.
EGFR p.T790M detected; variant allele fraction 12%.

VARIANT DETAILS				
Gene	Transcript	Exon	Coding change	Protein change
EGFR	NM_005228.5	21	c.2573T>G	p.L858R
EGFR	NM_005228.5	20	c.2369C>T	p.T790M

INTERPRETATION
Two EGFR substitutions are detected. Correlate with prior molecular testing and treatment history. The phase relationship of the two variants is not determined. Tumor-only testing does not establish whether a variant is somatic or germline.

QUALITY METRICS
DNA input: 42 ng | Mean target depth: 1540x
L858R locus depth: 1640x | T790M locus depth: 1240x
Target bases at >=250x coverage: 99.4%
All listed quality metrics are simulated for this illustration.

METHOD
Illustrative targeted DNA sequencing of EGFR exons 18-21. Single-nucleotide variants and small insertions/deletions assessed. Illustrative reporting threshold: 5% variant allele fraction. No RNA fusion or copy-number assessment is included.

LIMITATIONS
Low tumor content, sampling, and variants below the reporting threshold may limit detection. A negative result would not exclude alterations outside the interrogated regions. No paired normal sample was analyzed. Assay specifications are illustrative only.

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Lena Varnell | 48271639 | MP32-006381 | Page 1 of 1

MOLECULAR RESULT - EXACT EXCERPTS

Original synthetic report

EGFR p.T790M detected

Institution-developed pipeline

Patient name: ****
***** number: *****
Date of birth: **/**/****
Ordering provider: ****, MD
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Collected: **/**/**** 09:25
Received: **/**/**** 10:12
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MOLECULAR TESTING
Block A1 selected; estimated viable tumor cellularity 40%.
Targeted DNA sequencing reported separately as ****-*****.
The diagnosis is based on this biopsy; pathologic stage is not assigned.

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Institution-developed pipeline

Patient name: ****
***** number: *****
Date of birth: **/**/****
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MOLECULAR PATHOLOGY REPORT
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QUALITY METRICS
DNA input: 42 ng | Mean target depth: *****
***** locus depth: ***** | ***** locus depth: *****
Target bases at >=250x coverage: **.4%
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METHOD
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Institution-developed pipeline

EGFR p.***** detected

Refined local rules

Patient name: *****
***** number: *****
Date of birth: *****
Ordering provider: *****

Accession: *****

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Refined local rules

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Date of birth: *****
Ordering provider: *****

Accession: *****
Parent accession: *****

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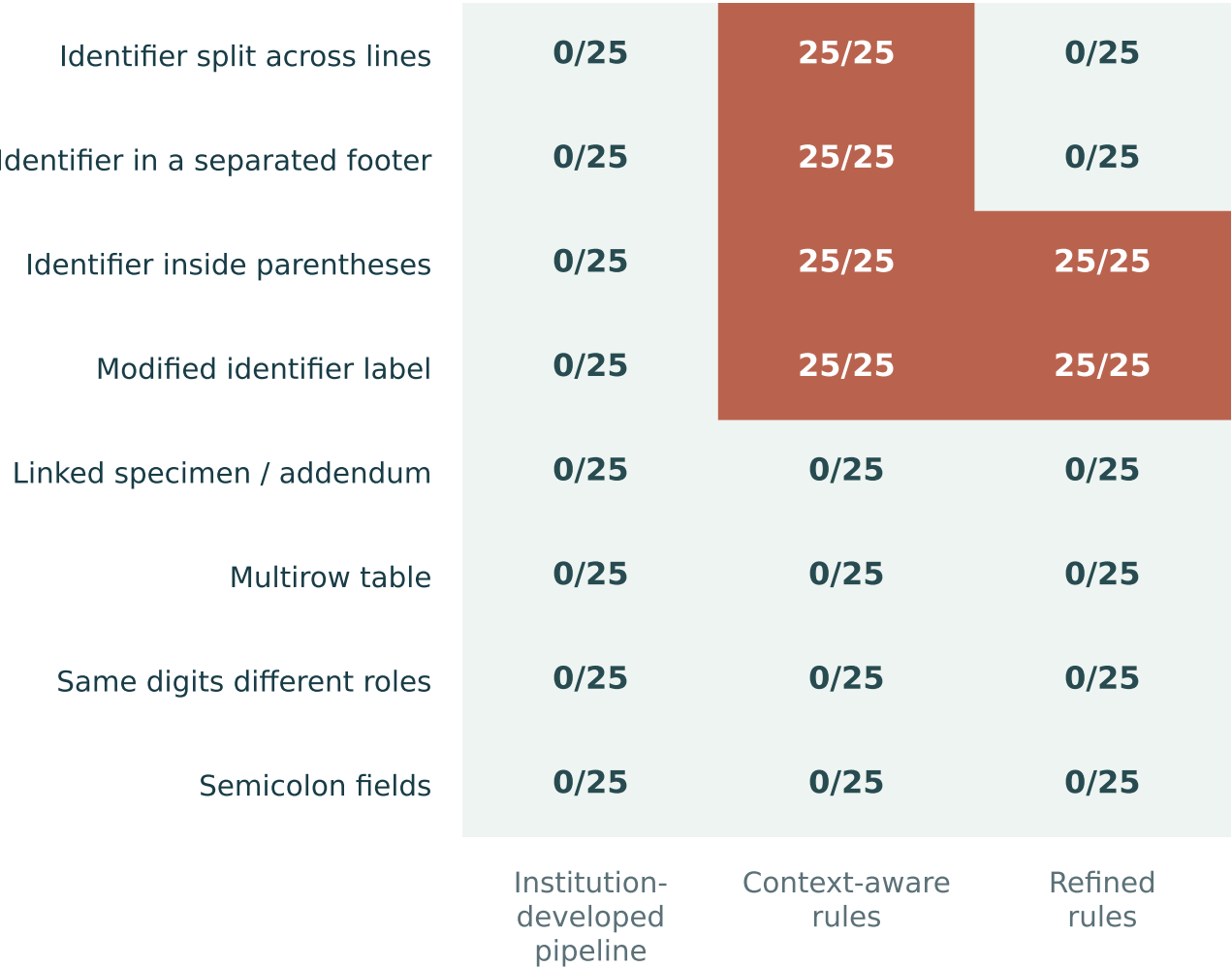
Electronically signed: ****, MD
***** | | ***** | Page 1 of 1

Refined local rules

EGFR p.T790M detected

Figure 2. Molecular information lost or preserved in a synthetic lung case. A fictional lung adenocarcinoma biopsy and linked EGFR report are shown before redaction and after processing by the institution-developed pipeline and refined local rules. The enlarged excerpt shows EGFR p.T790M masked by the pipeline and retained by the refined rules. All report details are invented. These are actual local-method outputs; no hosted LLM was run on this illustration. The case is excluded from both benchmark cohorts. Repeated synthetic banners and blank margins are omitted for display; source text and saved outputs remain unchanged.

A Identifier-control reports exposing tested digits (out of 25)



B Preservation and overall performance

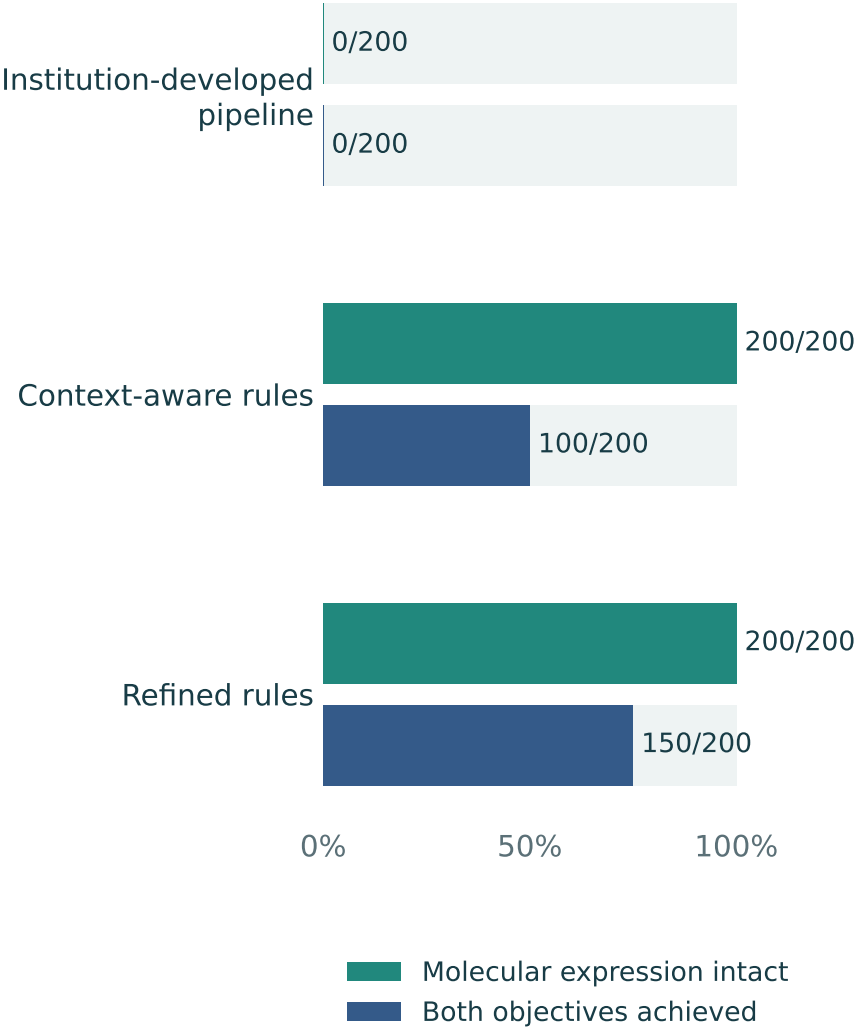
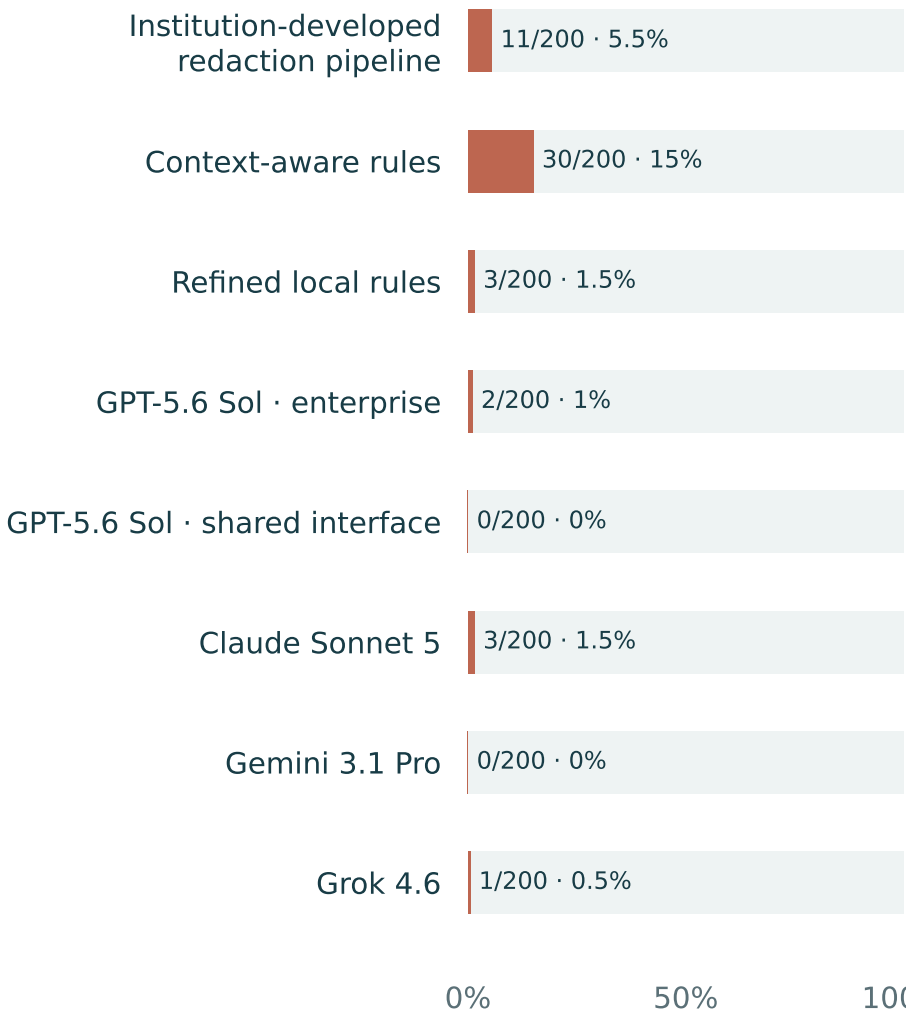


Figure 3. Designed report layouts reveal persistent local failures. This separate local experiment contains 200 unique synthetic pairs across eight investigator-designed report patterns. Patterns combine known local weaknesses and contextual challenges, including line breaks, parentheses, footer placement, tables, modified labels, and specimen links. Each pattern contains 25 pairs: 20 original and five additional distinct pairs. Panel A counts identifier-control reports exposing designated digits out of 25 controls per pattern; panel B summarizes molecular preservation and joint success across all 200 pairs. Refinement corrected two patterns but left parenthetical-field and modified-label failures, reducing exposed controls from 100 to 50. Improvement remained inconclusive when grouped by pattern (p=.50). These cases were not pooled with the main comparison.

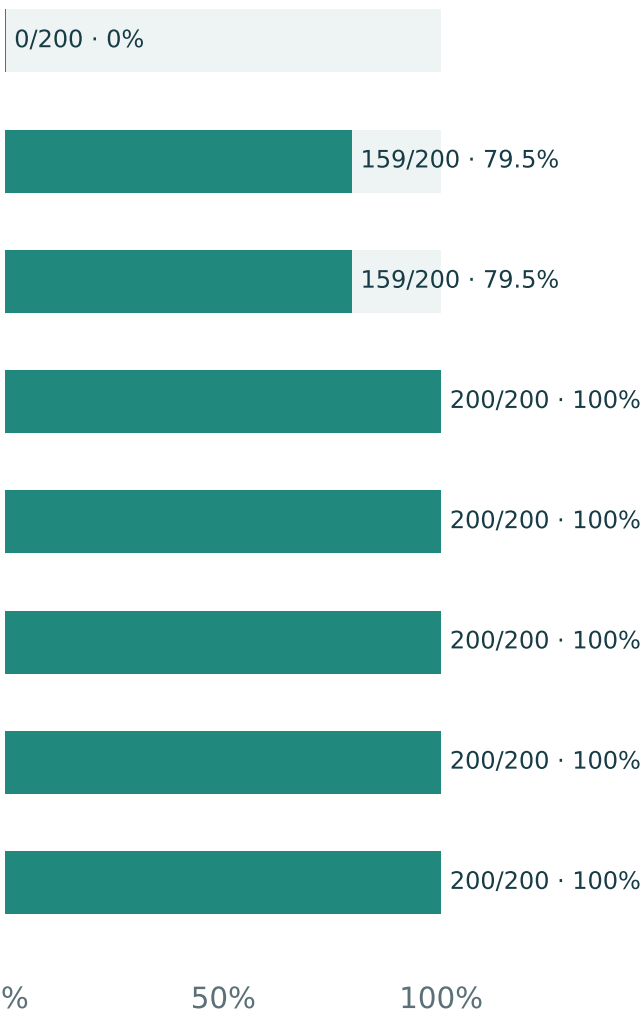
A Tested identifying digits visible

Lower is better



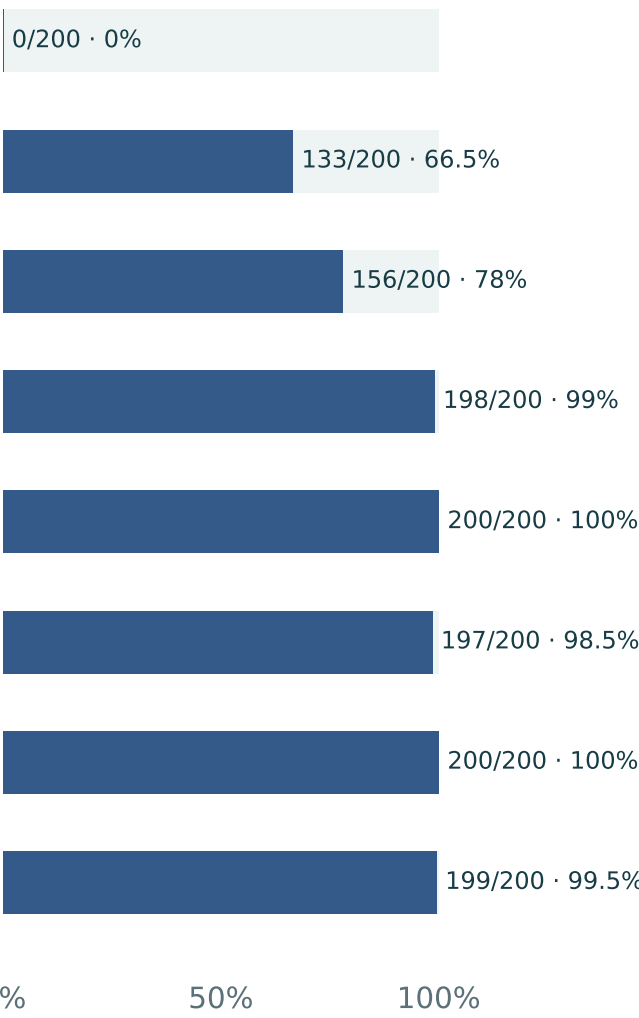
B Molecular expression intact

Higher is better



C Both objectives achieved

Higher is better



Supplementary Figure S1. Identifier removal, molecular preservation, and joint success. All eight approaches were evaluated on the same 200 unique report pairs. Panel A counts controls retaining tested identifying digits; panel B counts molecular reports retaining the complete tested expression; panel C requires both outcomes to succeed within a pair. Zero tested-digit exposure does not establish removal of all identifiers. Five LLM configurations represent four model families, with GPT evaluated through Enterprise and the shared interface used by the other models. Refined rules were evaluated subsequently without changes to their implementation.